DRAGOON MOUNTAINS GROWING GUIDES

Hydroponic Nutrients Guide

Locally-Sourced Masterblend-Style Recipes · Dragoon Mountains, Arizona

Dragoon Mountains Growing Guides

Contents

Figure 1 – The Dragoon Mountains rising above the Sulphur Springs Valley, Cochise County, Arizona.

This is the locally-sourced companion to the Dragoon Mountains library. Pair it with:

- **A — Pepper Growing Guide** — the crop this programme was tuned around
- **B — Bato Bucket Hydroponic Setup** — the recommended build
- **D — A Hydroponic Primer** — climate, systems, seasonal calendar
- **E — Modern Masterblend Guide** — the purchased equivalent of every recipe below
- **G — pH Up & pH Down Replacements** — local acids and alkalis for this programme

How to use this guide

Everything below targets the same nutrient profile as the purchased three-part Masterblend programme in **Guide E**. The difference is where the raw materials come from: oak ash instead of a bag of 4-18-38; bat guano instead of calcium nitrate-supplied N; caliche dissolved in vinegar instead of a scoop of calcium nitrate; epsomite crust instead of an Epsom-salt bag.

You can run the locally-sourced programme end-to-end, or you can blend it — for example, purchased Masterblend + local caliche calcium + local epsomite. The nutrient ppm targets don't change either way.

The EC meter and pH meter are not optional. Local brews vary from batch to batch, and the only way to know what you actually dosed is to measure. Target EC 1.8–2.4 mS/cm and pH 5.8–6.2 for most fruiting crops.

1. Target Nutrient Profile

This is the same profile as the purchased programme in Guide E. Match it, and the crop doesn't know the difference.

Macronutrient	Target ppm	Micronutrient	Target ppm
Nitrogen (N)	116	Iron (Fe, chelated)	2.4
Phosphorus (P)	47	Boron (B)	1.2
Potassium (K)	188	Manganese (Mn)	1.2
Calcium (Ca)	113	Zinc (Zn)	0.3
Magnesium (Mg)	41	Copper (Cu)	0.3
Sulfur (S)	58	Molybdenum (Mo)	0.06

The three parts, mapped to local sources

Part	Purchased (Guide E)	Local equivalent
A — K, P, N + micronutrients	Masterblend 4-18-38 (12 g / 5 gal)	Oak-ash potash + aerated bat-guano tea + chelated iron + micronutrient micro-doses
B — Mg + S	Epsom salt (6 g / 5 gal)	Epsomite crust from Sulphur Springs Valley
C — Ca + N	Calcium nitrate (12 g / 5 gal)	Caliche dissolved in apple-cider vinegar + nitrified guano tea

Never pre-combine Parts A, B and C as concentrates — they will react and precipitate, exactly as the purchased equivalents do. Dissolve each separately into the reservoir.

2. Part A — Potassium, Phosphorus, Nitrogen & Micronutrients

This is the recipe that replaces the 4-18-38 bag.

Nutrient	Local source	Production method
Potassium (K)	Oak wood ash (abundant in the Dragoons)	Burn seasoned oak hardwood. Steep ash in water 24 hrs. Strain through cloth. Use the liquid potash extract.
Phosphorus (P)	Bat guano (Cochise County caves)	Collect guano. Steep 1 cup per gallon of water. Aerate vigorously 24–48 hrs. Strain finely.
Nitrogen (N)	Aerated guano tea	Continued aeration converts ammonium (NH_4^+) to nitrate (NO_3^-) via nitrifying bacteria. Use within 48 hrs.
Iron (Fe), chelated	Red granite clay + oak bark / leaves	Brew strong oak-bark tea. Soak iron-rich red clay in it 48 hrs. Strain. The tannins chelate the iron.
Copper (Cu)	Bisbee malachite / azurite	Dissolve a small chip in dilute apple-cider vinegar. Dilute 1:100 before reservoir use.

Nutrient	Local source	Production method
Boron (B)	Desert caliche, Sulphur Springs Valley soil	Boron concentrates naturally in AZ arid soils. Brew a strong caliche-soil tea. Use sparingly.

Oak is the anchor species for local nutrient brewing in the Dragoons. Potassium lives in the ash, chelating tannins live in the bark and leaves, and both have been used across the Southwest for centuries. If you have an oak woodlot, you have Part A.

3. Part B — Calcium

Caliche (calcium carbonate hardpan) is abundant throughout the Sulphur Springs Valley east of the Dragoons. To make plant-available calcium:

1. Collect caliche chunks from exposed hardpan layers.
2. Dissolve slowly in apple-cider vinegar (from your fermentation set-up) — this produces calcium acetate, which is fully plant-available.
3. Combine with nitrified guano tea to provide calcium and nitrogen together — the same pairing the purchased calcium nitrate offers.
4. Alternative: dolomite limestone dissolved in dilute acid provides both Ca and Mg simultaneously.

4. Part C — Magnesium & Sulfur

The Sulphur Springs Valley is named for its mineral deposits. Look for **epsomite** — white crystalline crusts around dry springs and mineral seeps. Epsomite is magnesium sulphate in its natural form, identical to commercial Epsom salt.

Alternative: collect sulphur-rich spring water and dissolve dolomite into it.

The pepper Mg bump applies here too. Peppers are heavy magnesium users. Once the first fruit is sizing, increase your epsomite or Epsom-salt dose to the equivalent of **2.4 g Epsom per gallon** and hold through harvest. This mirrors the purchased-programme rule in Guide E Section 6, and is the single most common deviation from the base recipe.

5. Micronutrient Production

Nutrient	Local source	Method
Iron (Fe)	Red granite clay + oak bark	Steep iron-rich clay in oak tannin tea 48 hrs; tannins chelate it into plant-available form.
Manganese (Mn)	Dark metamorphic rock, streambed sediment	Collect mineral-stained streambed mud. Brew as dilute tea. Strain before use.
Zinc (Zn)	Tombstone / Bisbee zinc-bearing ores	Dissolve a zinc-ore chip in dilute acid. Heavily dilute before adding to reservoir.
Copper (Cu)	Bisbee copper ores (malachite, azurite)	Dissolve a small piece in dilute apple-cider vinegar. Tiny amounts only — Cu is toxic at high doses.
Boron (B)	Desert caliche, dry lakebeds, borax deposits	Use Sulphur Springs Valley soil tea. Use very sparingly.
Molybdenum (Mo)	Wild legume root nodules (mesquite, clover, palo verde)	Ferment root nodules in water. Provides trace Mo — the hardest element to source locally.

Micronutrient safety. Copper and boron are toxic to plants at high concentrations. Always make a very dilute stock solution and add drop by drop, testing EC between additions. Zinc and manganese follow the same rule — small amounts, dilute, measure.

6. Feasibility Summary

Component	Locally achievable?	Notes
Potassium (K)	✓ Yes	Oak wood ash is an excellent source.
Phosphorus (P)	✓ Yes	Bat guano from Cochise County caves.
Calcium (Ca)	✓ Yes	Caliche is ubiquitous in the area.
Magnesium (Mg)	✓ Likely	Hunt for epsomite deposits in Sulphur Springs Valley.
Sulfur (S)	✓ Yes	The valley is named for its sulphur deposits.

Component	Locally achievable?	Notes
Iron (Fe), chelated	✓ Yes	Oak tannin + red clay method is effective.
Copper (Cu)	✓ Yes	Bisbee copper ores are world-class.
Boron (B)	✓ Yes	AZ desert soils concentrate boron naturally.
Nitrogen (nitrate form)	■ Partial	Hard to get pure nitrate — aerated guano tea helps.
Mn, Zn (trace)	■ Difficult	Available but hard to dose accurately.
Molybdenum (Mo)	✗ Very hard	Legume tea only provides trace amounts.

A practical split for most growers: use **local Parts B and C** (epsomite, caliche, guano, oak ash) and purchase a **low-cost micronutrient mix** for the trace elements (Mo, Zn, Mn, B). You get most of the self-sufficiency at a fraction of the accuracy risk.

7. Recommended Workflow

Step	Action	Source
1	Brew oak-ash potash extract	Local oaks → burn → steep
2	Prepare aerated bat-guano tea	Cochise County bat roosts
3	Make caliche-vinegar calcium extract	Sulphur Springs Valley caliche
4	Brew oak-tannin + red-clay iron chelate	Local granite hillsides
5	Collect epsomite or use purchased Epsom salt	Local mineral seeps OR purchase
6	Make copper micro-dose (Bisbee malachite)	Bisbee-area copper ores
7	Brew desert-soil boron tea	Sulphur Springs caliche soil
8	Combine all parts separately into reservoir — check EC and pH	EC + pH meters (purchased)

Regardless of how nutrients are sourced, **an EC meter and a pH meter are essential**. Target EC 1.8–2.4 mS/cm and pH 5.8–6.2 for most fruiting crops. These instruments cannot be replaced with local materials.

For the acid and alkali recipes that complete this programme — prickly-pear vinegar for daily pH down, oak-ash lye for pH up, fermented citric acid for the bicarbonate scrub — see **Guide G: pH Up & pH Down Replacements**.

Closing

Cochise County is one of the richest mineral landscapes in the American Southwest. Oak for potassium, bat caves for phosphorus, caliche for calcium, epsomite for magnesium, Bisbee copper and Tombstone zinc for traces — the same geology that made this a mining frontier makes it an unusually self-sufficient place to grow food hydroponically.

The purchased three-part Masterblend programme in Guide E is faster and more precise. The locally-sourced programme here takes longer, drifts more, and asks more of your measuring kit — but it costs almost nothing to run, and it ties the reservoir back into the land it sits on. Either gets a pepper to ripen. Most growers end up somewhere in the middle: a bag of Masterblend for the base, and a brew of caliche calcium or an epsomite crust for the parts the valley gives you for free.